Mortality Equations for CONUS

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Data

We took the FIA mortality data subset, filtered out missing data (primarily crown ratio and heights). This left 4343197 observations.

The observations are distributed among states is show in Table 1.

Table 1: Mortality Observations by State

State	n
MN	396255
ME	327368
WI	321417
MI	298910
OR	204776
GA	176540
NY	170537
PA	152055
AL	141930
WA	137839
VA	136219
MO	135853
NC	124516
AR	112689
SC	112014
MS	111427
TN	109749
CA	105913
MT	95025
KY	88127
TX	85291

Table 1: Mortality Observations by State

State	n
CO	79722
WV	78684
NH	73331
ID	61960
OH	53553
VT	52214
FL	49165
LA	43621
IN	42858
OK	36910
IL	30733
MA	24706
UT	24542
IA	17511
AZ	16740
NM	15355
SD	13697
NJ	13336
MD	13040
WY	11708
CT	10842
KS	10777
NE	7360
ND	5496
RI	4712
DE	4444
NV	1730

Base Rate Mortality

Daniel Kanneman and Amos Tversky[1] made the point that any putative model should improve upon the *base rate*. In our case, the base rate is the average mortality rate for each species accounting for no other factors. We calculate this by taking the annualized mean mortality rate for each species in the data set.

The base rate estimates for 183 species are distributed as shown in Figure 1. Most base rate annual survival estimates are between 98 and 100%.

[1] Kahneman D. and Tversky, A. (1973). On the Psychology of Prediction. Psychology Review. 80(4), 237-251. doi: 10.1037/h0034747

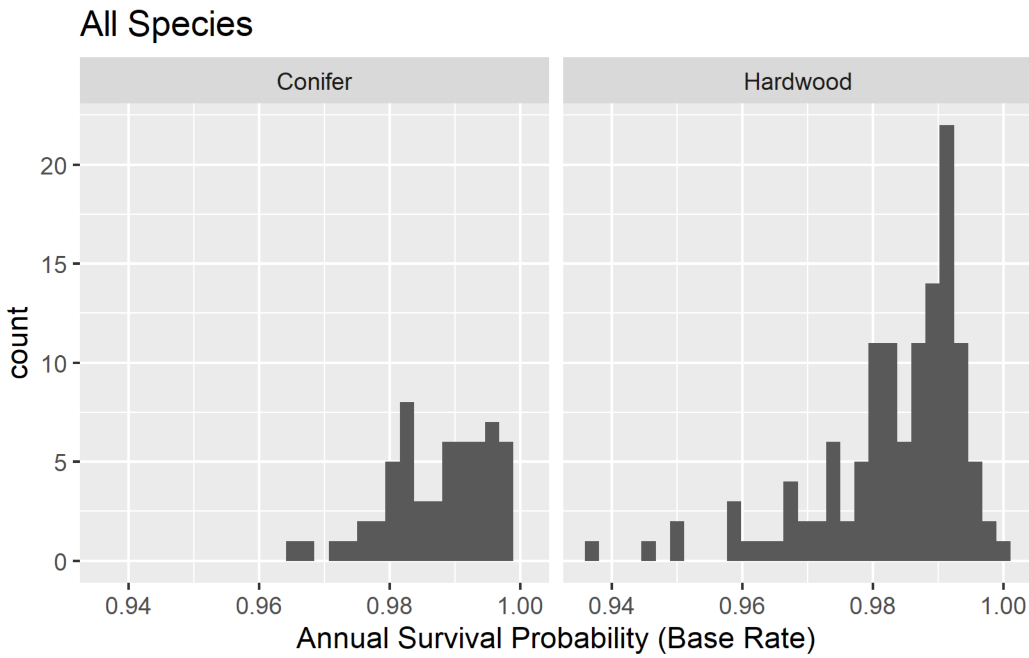
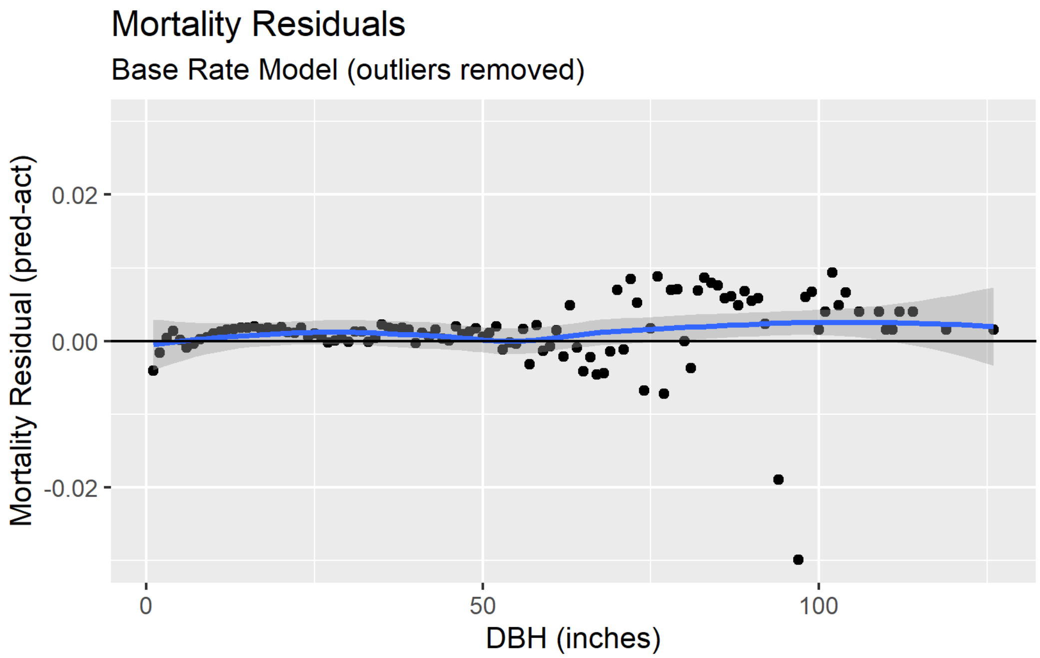


Figure 1

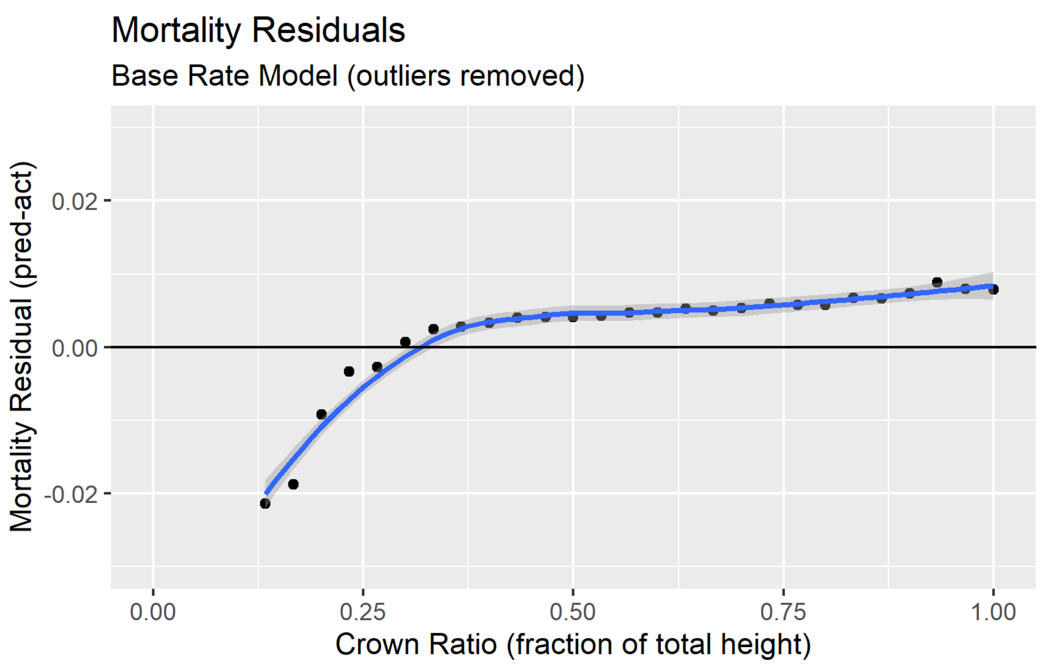
Base Rate Residuals

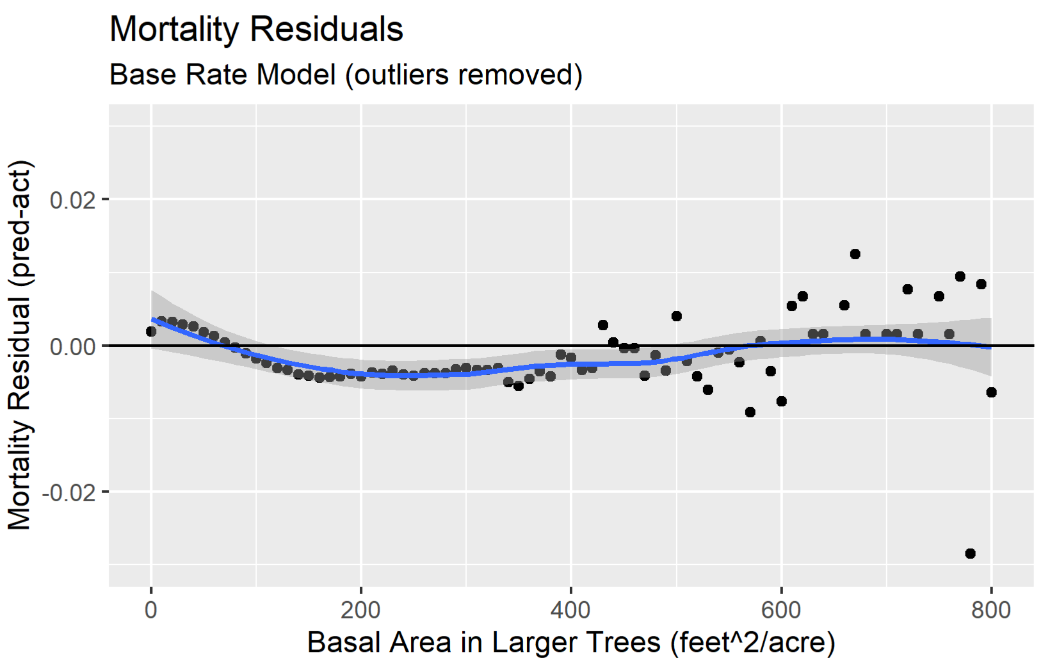
The following residual graphs show the average errors for all species over classes of possible explanatory variables.

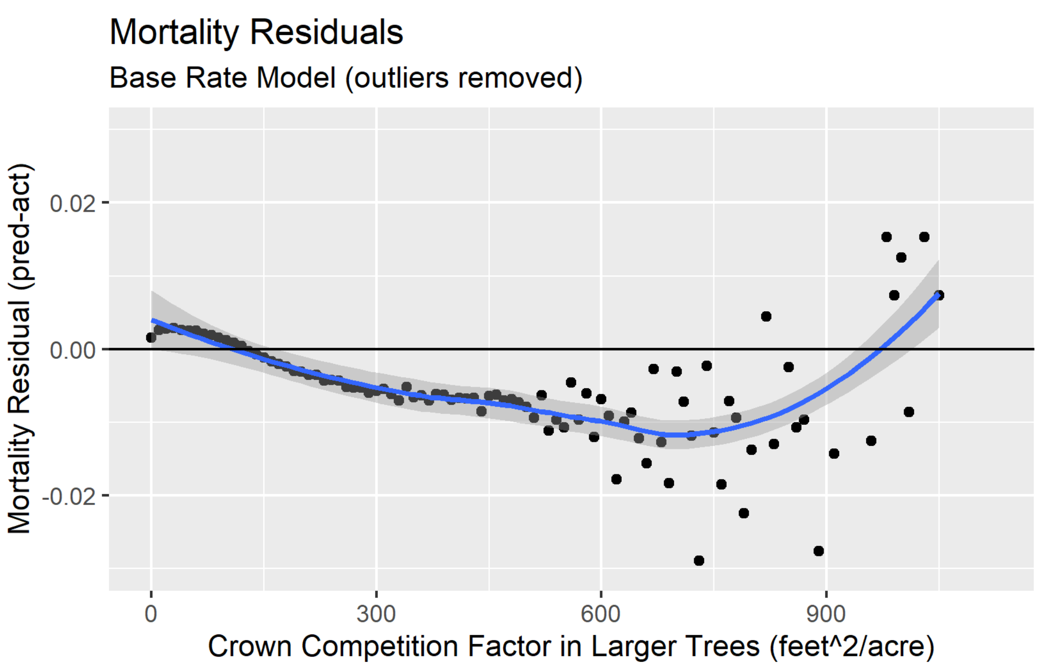

Mortality Residuals
Base Rate Model (outliers removed)
Mortality Residual (pred-act)
0.02
0.00
-0.02
0
50
100
DBH (inches)

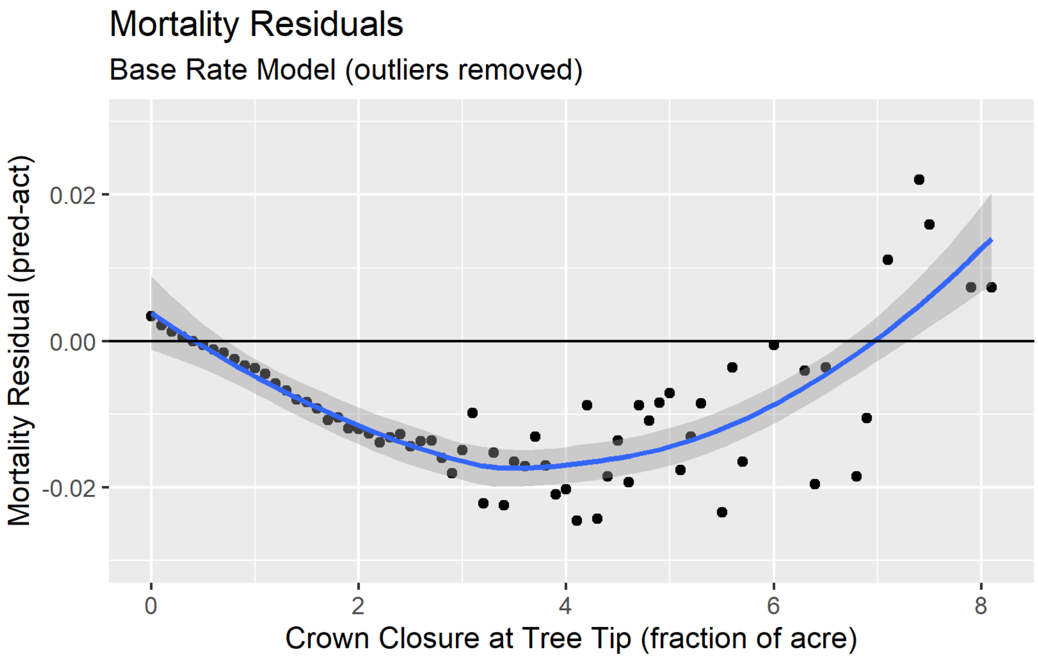

Mortality Residuals
Base Rate Model (outliers removed)
Mortality Residual (pred-act)
0.02
0.00
-0.02
0
100
200
300
Height (feet)

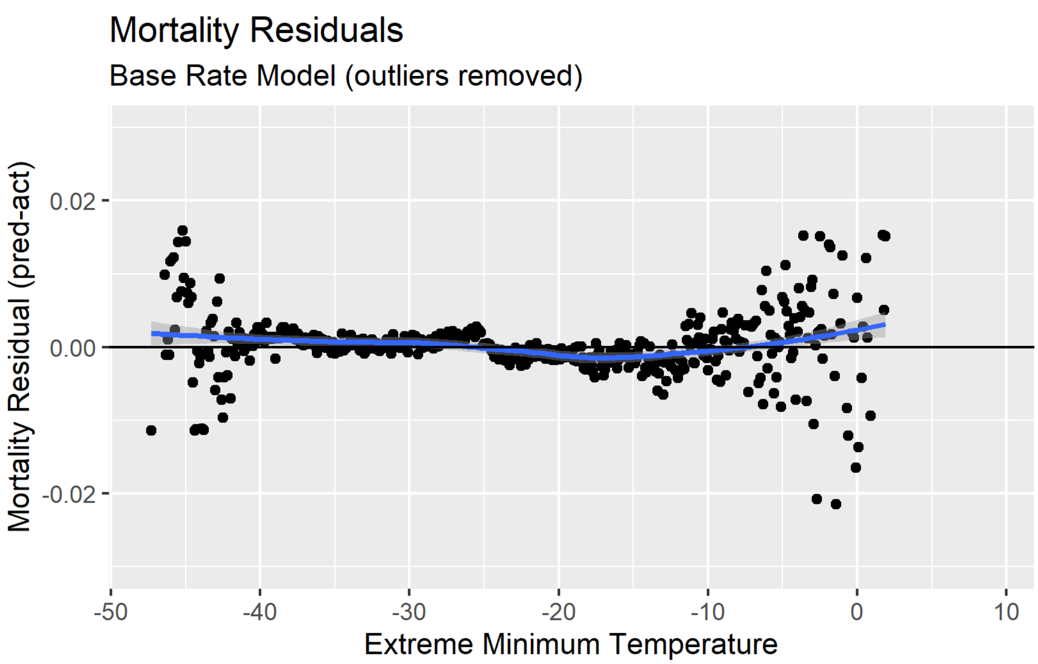

Mortality Residuals
Base Rate Model (outliers removed)
Mortality Residual (pred-act)
0.02
0.00
-0.02
0
10
20
Crown Length/DBH (feet/inches)

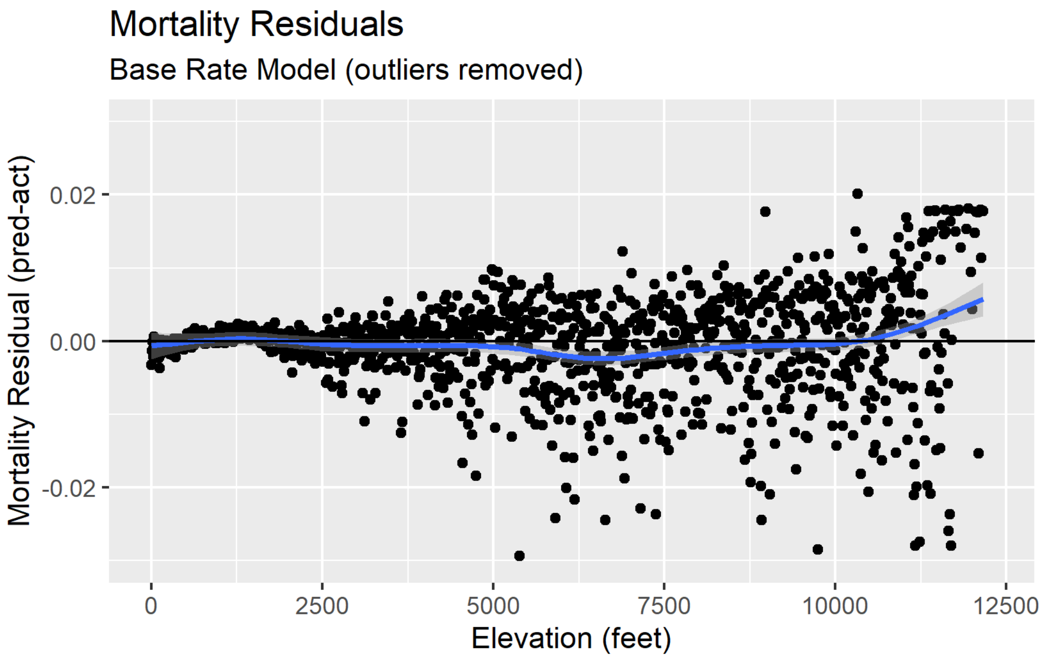

Mortality Residuals
Base Rate Model (outliers removed)
Mortality Residual (pred-act)
0.02
0.00
-0.02
0.00
0.25
0.50
0.75
1.00
Crown Ratio (fraction of total height)

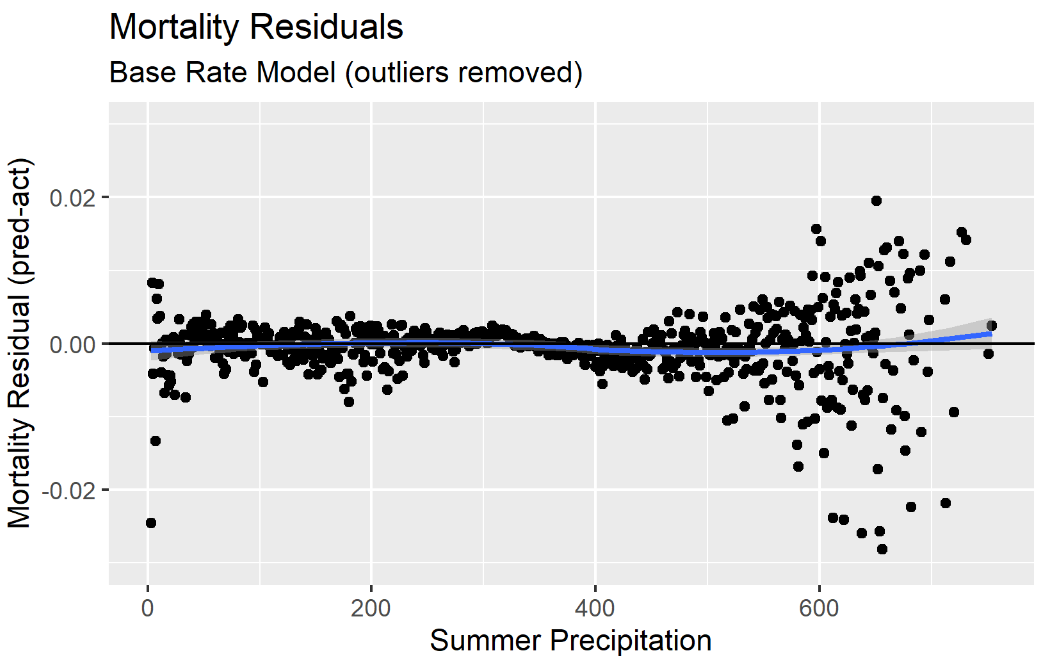

Mortality Residuals
Base Rate Model (outliers removed)
Mortality Residual (pred-act)
0.02
0.00
-0.02
0
200
400
600
800
Basal Area in Larger Trees (feet^2/acre)

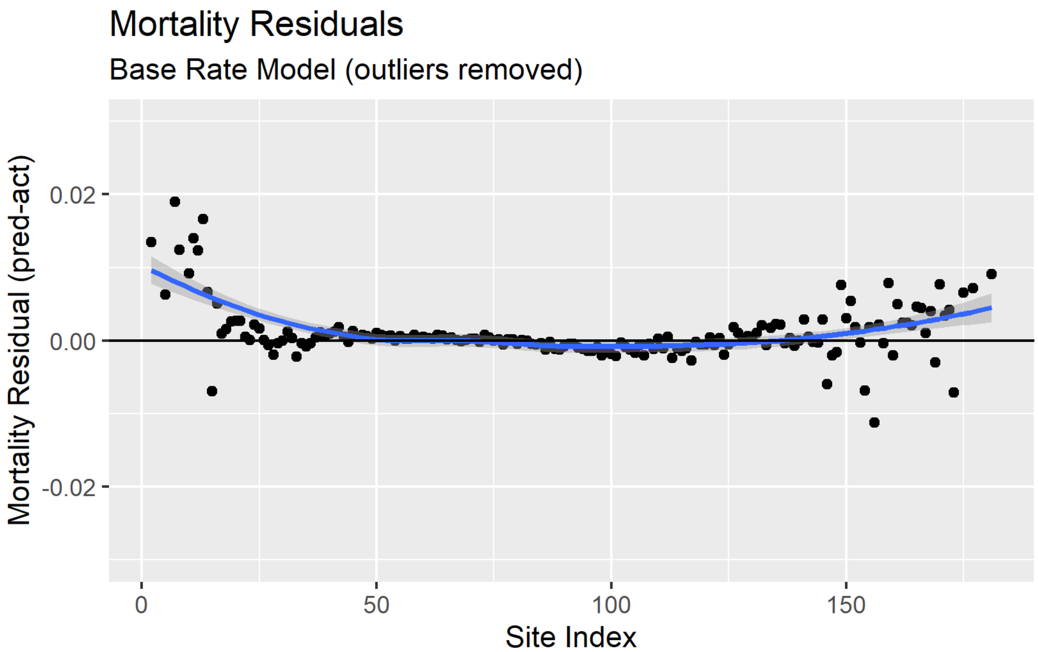

Mortality Residuals
Base Rate Model (outliers removed)
Mortality Residual (pred-act)
0.02
0.00
-0.02
0
300
600
900
Crown Competition Factor in Larger Trees (feet^2/acre)

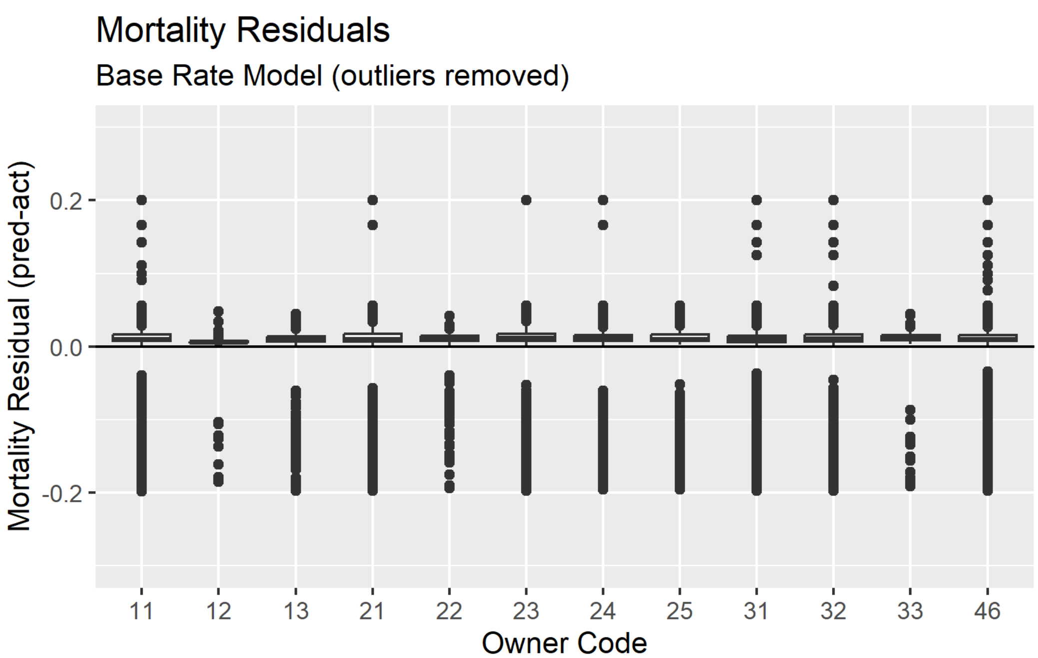

Mortality Residuals
Base Rate Model (outliers removed)
Mortality Residual (pred-act)
0.02
0.00
-0.02
0
2
4
6
8
Crown Closure at Tree Tip (fraction of acre)

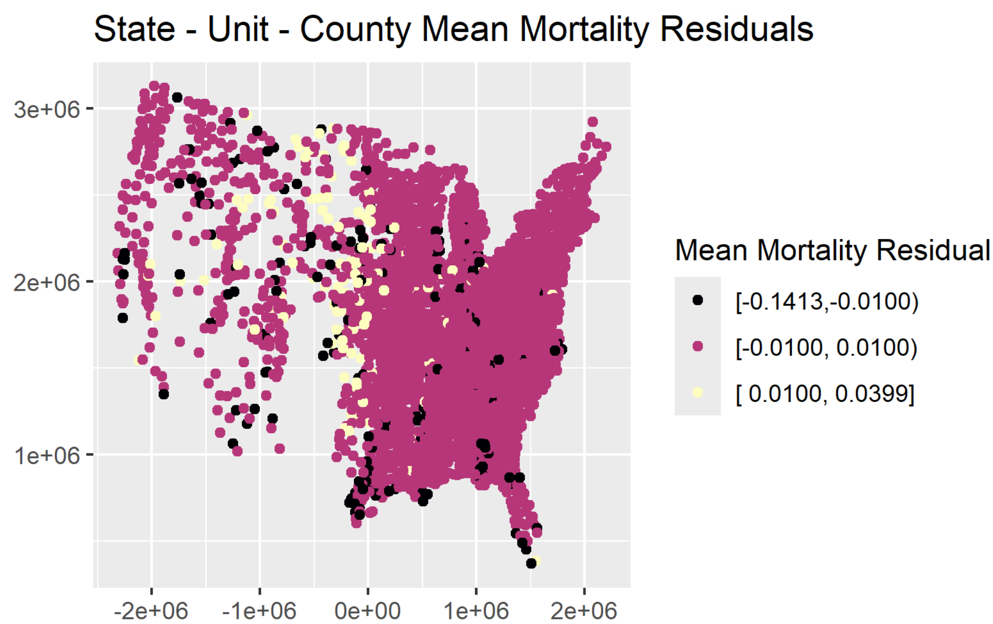

Mortality Residuals
Base Rate Model (outliers removed)
Mortality Residual (pred-act)
0.02
0.00
-0.02
-50
-40
-30
-20
-10
0
10
Extreme Minimum Temperature


Mortality Residuals
Base Rate Model (outliers removed)
Mortality Residual (pred-act)
0.02
0.00
-0.02
0
2500
5000
7500
10000
12500
Elevation (feet)


Mortality Residuals
Base Rate Model (outliers removed)
Mortality Residual (pred-act)
0.02
0.00
-0.02
0
200
400
600
Summer Precipitation


Mortality Residuals
Base Rate Model (outliers removed)
Mortality Residual (pred-act)
0.02
0.00
-0.02
0
50
100
150
Site Index


Mortality Residuals
Base Rate Model (outliers removed)
Mortality Residual (pred-act)
0.2
0.0
-0.2
11
12
13
21
22
23
24
25
31
32
33
46
Owner Code



Proposed Model

The model we propose to explain more of the mortality variation over the base rate is a Gompit function:

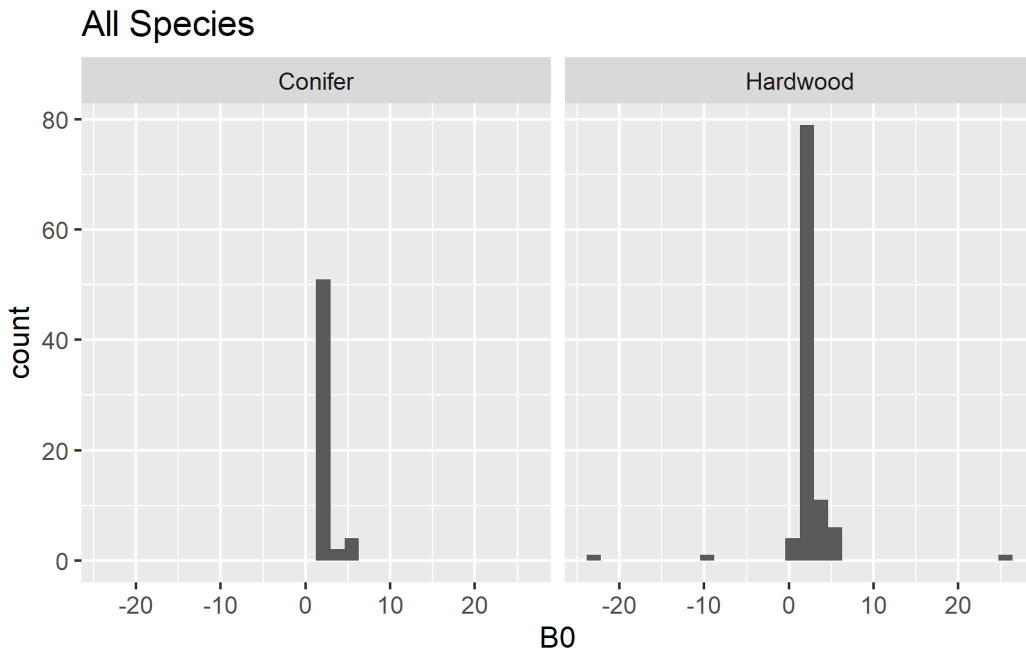
$$P_{surv} = 1 - e^{(-e^{(\beta_0 + \beta_1 (cr+0.01)^{\beta_2} + \beta_3 cch^{\beta_4})})} \tag{1}$$

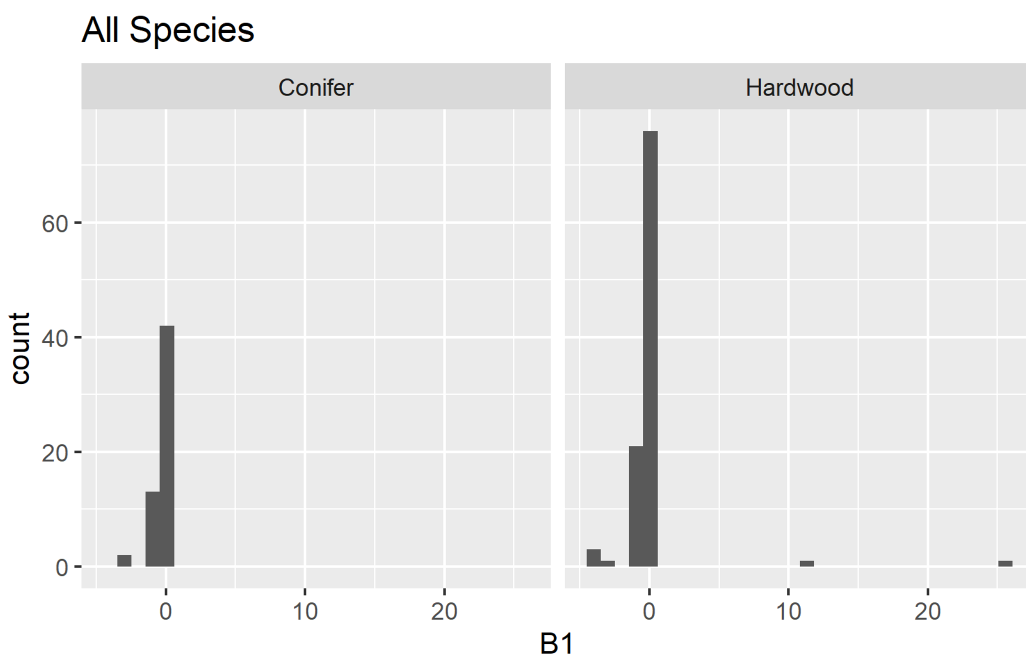
where:

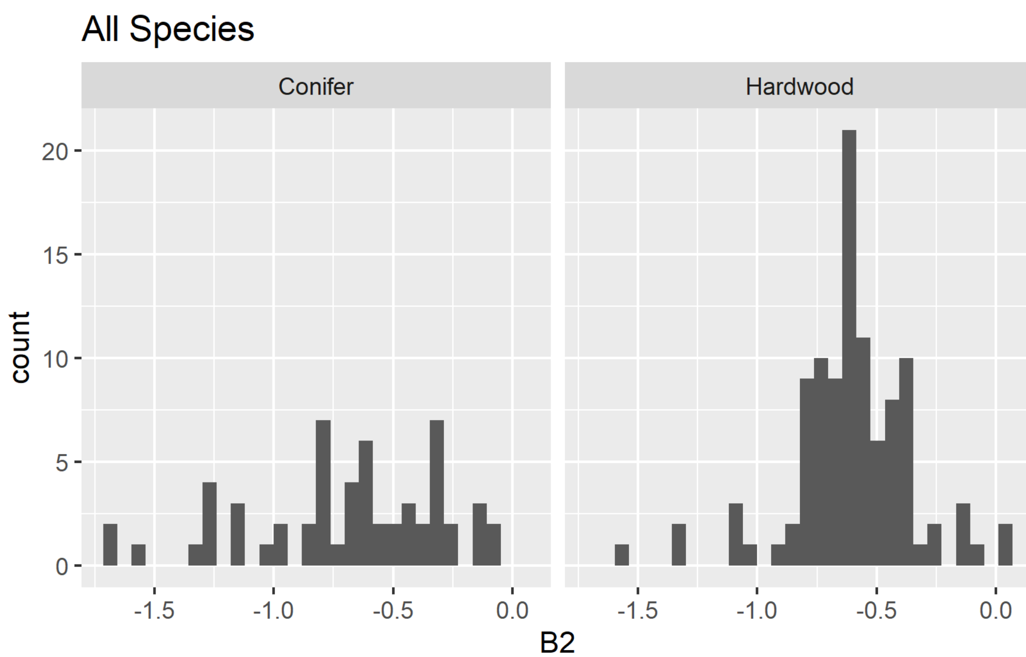
- P_{surv} : probability of survival
- cr : crown ratio
- cch : crown closure at tree tip

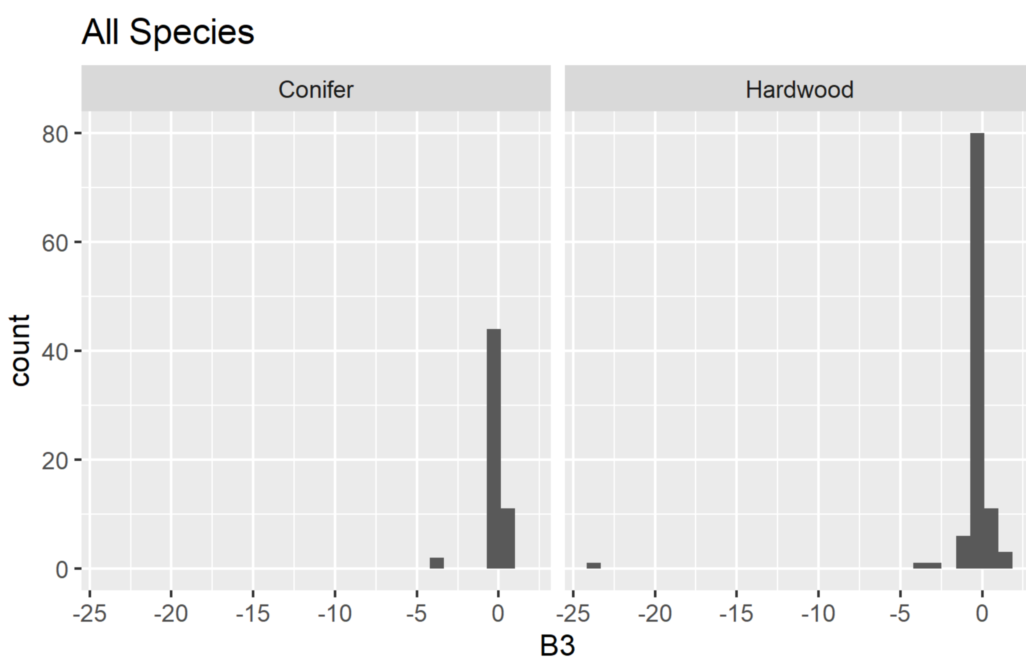
We used an integrative fitting approach to account for varying period lengths and we minimized the negative log-likelihood for each species separately.

The distributions of the parameter estimates for all species are shown below.


All Species
Conifer
Hardwood
count
B0
80
60
40
20
0
-20
-10
0
10
20


All Species
Conifer
Hardwood
count
B1
60
40
20
0
10
20


All Species
Conifer
Hardwood
count
20
15
10
5
0
-1.5
-1.0
-0.5
0.0
B2


All Species
Conifer
Hardwood
count
80
60
40
20
0
-25
-20
-15
-10
-5
0
B3

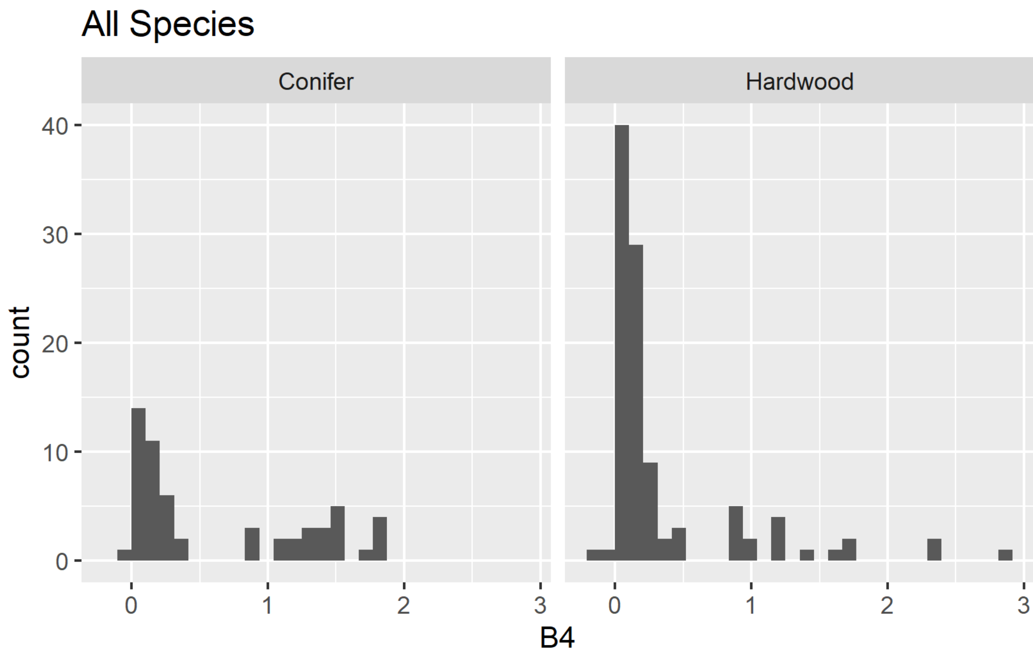


Figure 2 shows that all species' negative log likelihoods were reduced with the proposed model.

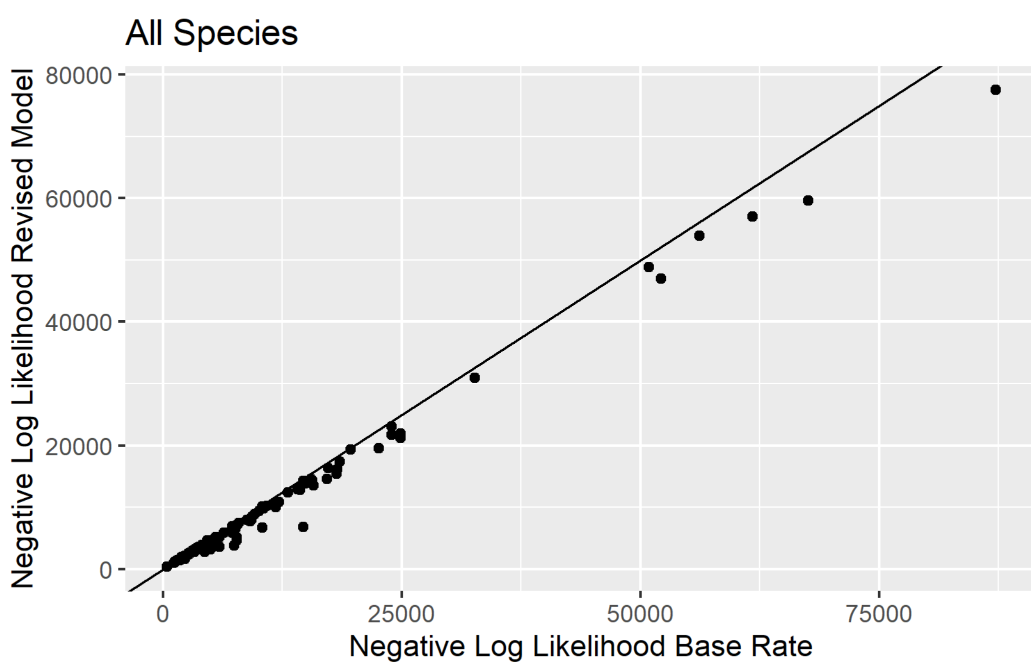
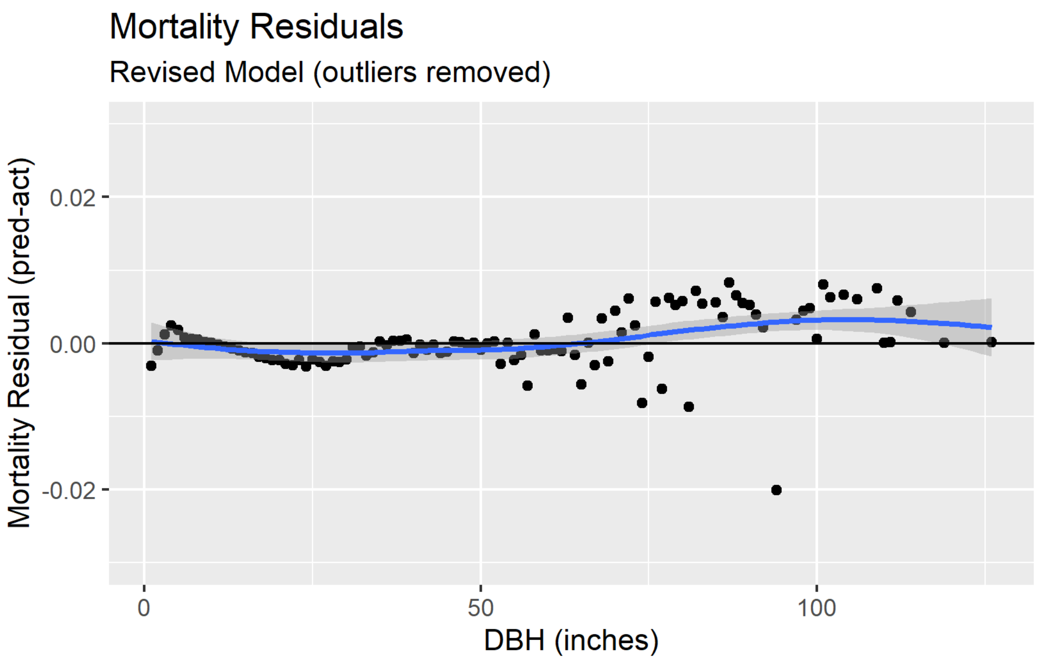
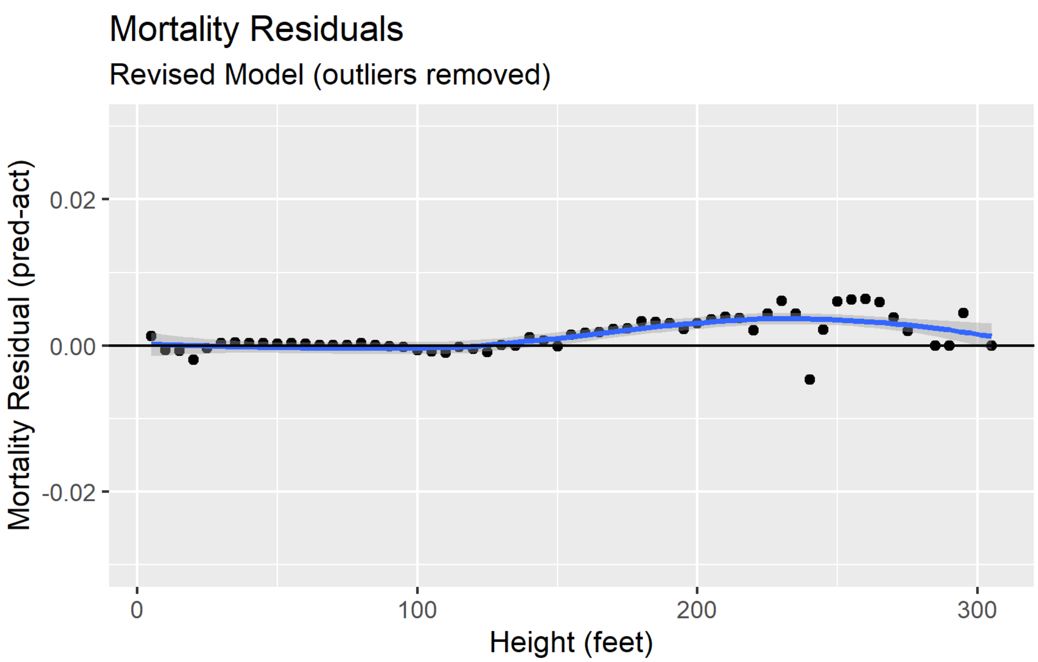


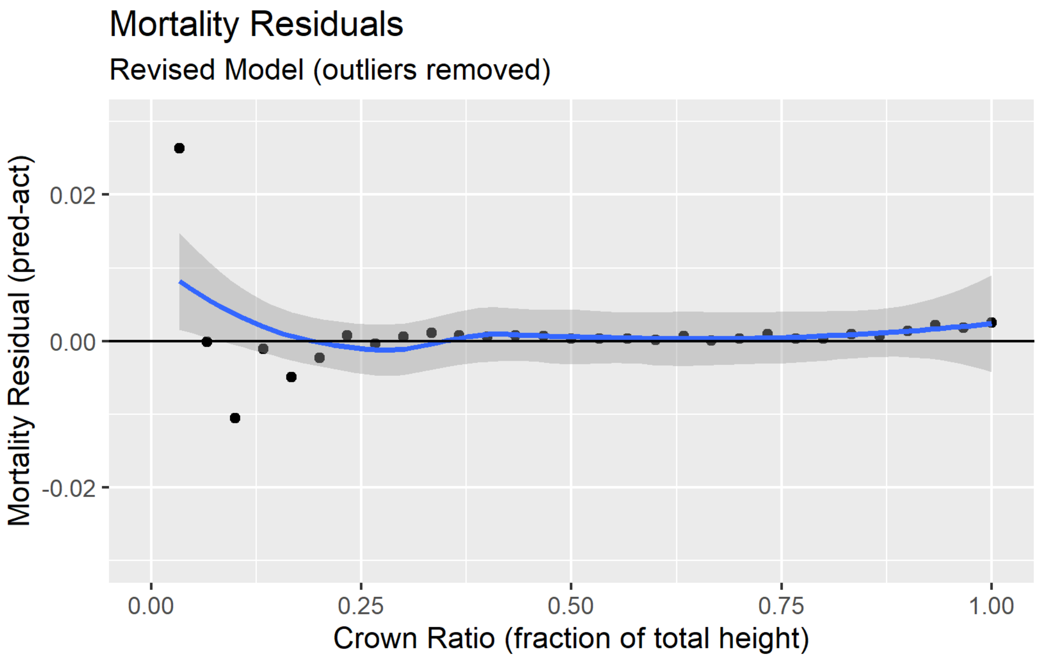
Figure 2

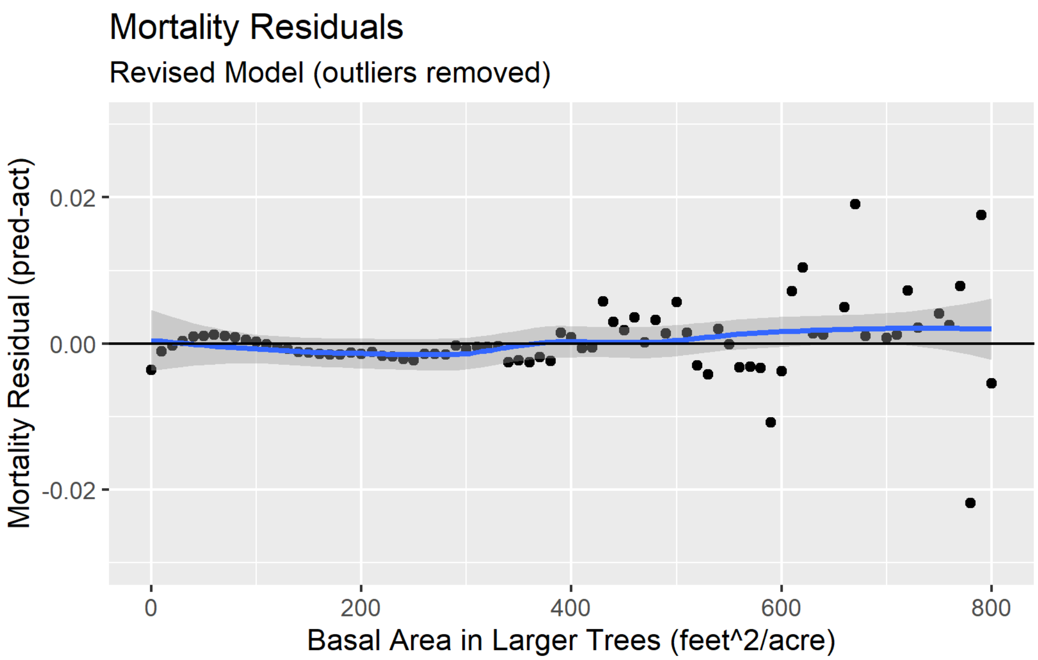
Proposed Model Residuals

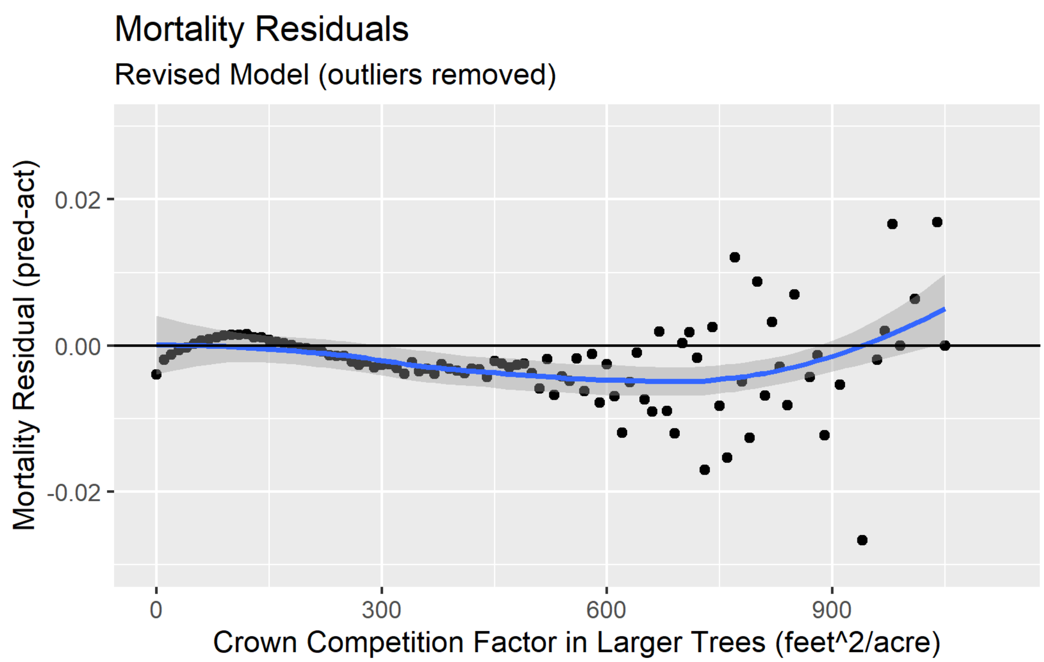
The following residual graphs show the average errors for all species where the fit improved over the base rate over classes of possible explanatory variables.






Mortality Residuals
Revised Model (outliers removed)
Mortality Residual (pred-act)
0.02
0.00
-0.02
0.00
0.25
0.50
0.75
1.00
Crown Ratio (fraction of total height)


Mortality Residuals
Revised Model (outliers removed)
Mortality Residual (pred-act)
0.02
0.00
-0.02
0
200
400
600
800
Basal Area in Larger Trees (feet^2/acre)


Mortality Residuals
Revised Model (outliers removed)
Mortality Residual (pred-act)
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-0.02
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300
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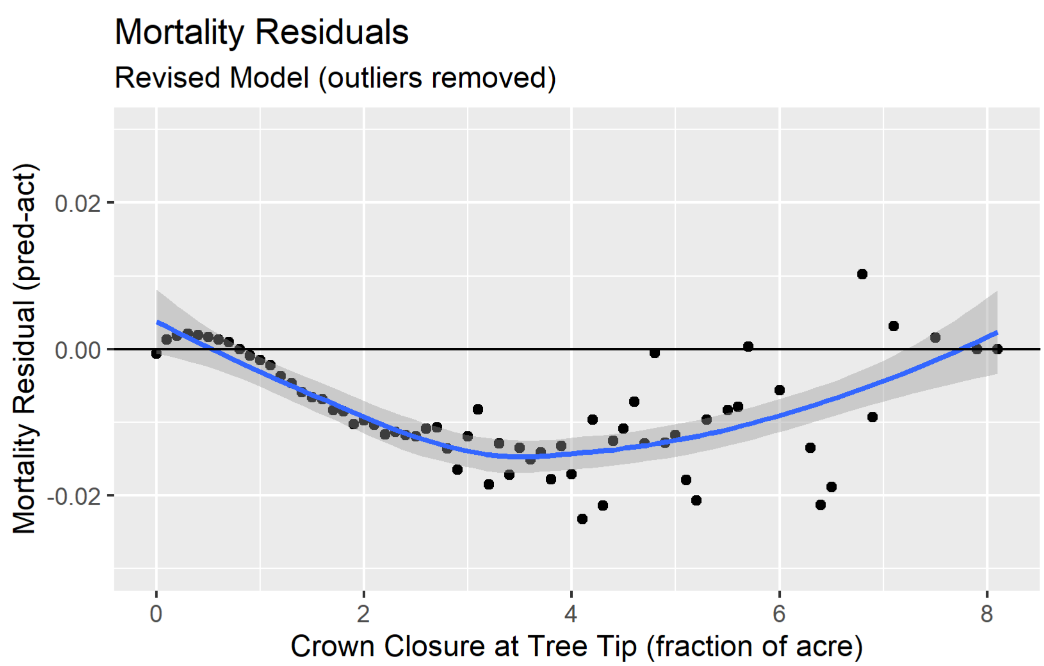
900
Crown Competition Factor in Larger Trees (feet^2/acre)


Mortality Residuals
Revised Model (outliers removed)
Mortality Residual (pred-act)
0.02
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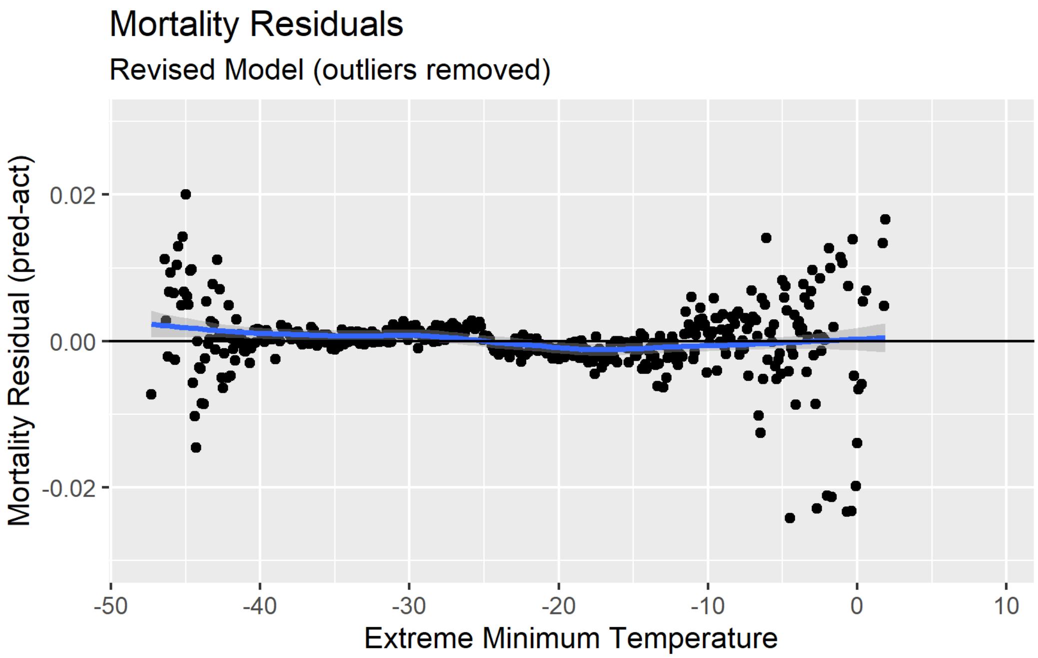
Crown Closure at Tree Tip (fraction of acre)


Mortality Residuals
Revised Model (outliers removed)
Mortality Residual (pred-act)
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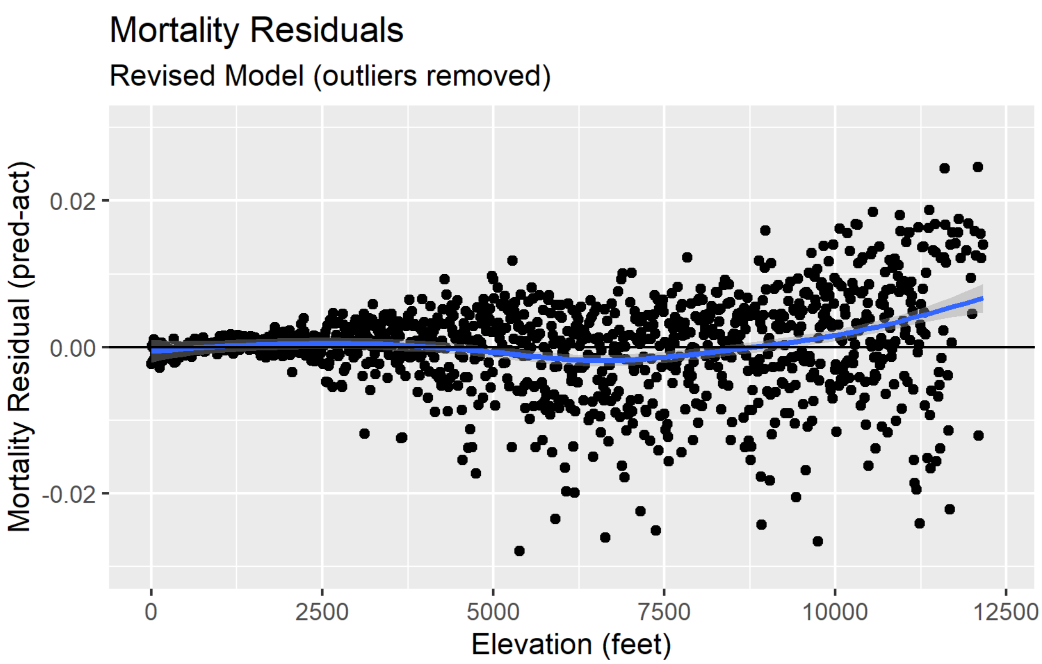
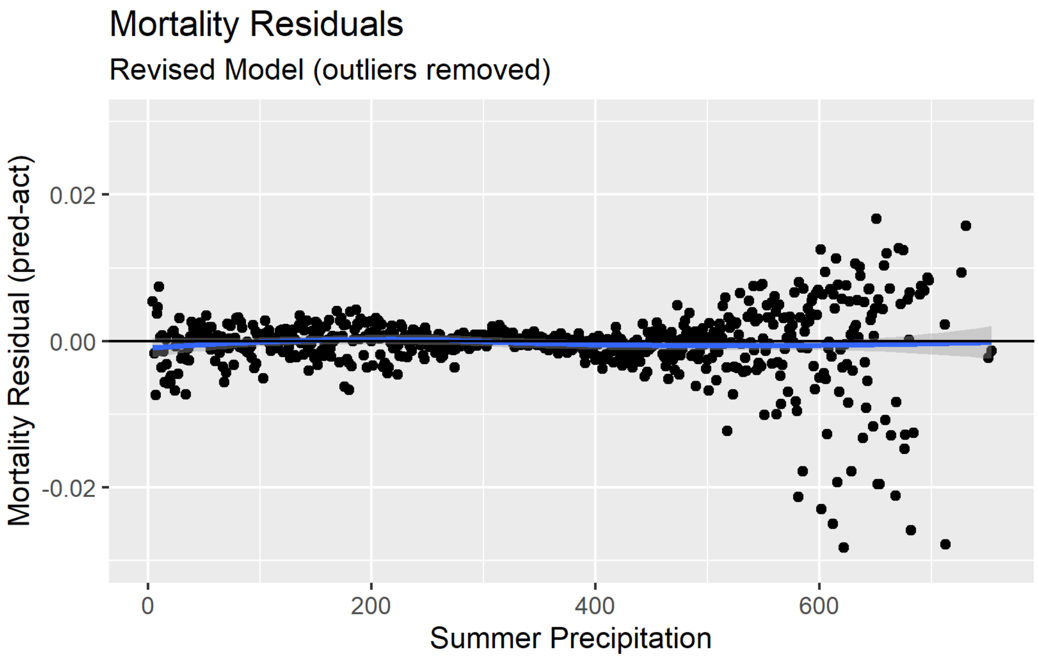
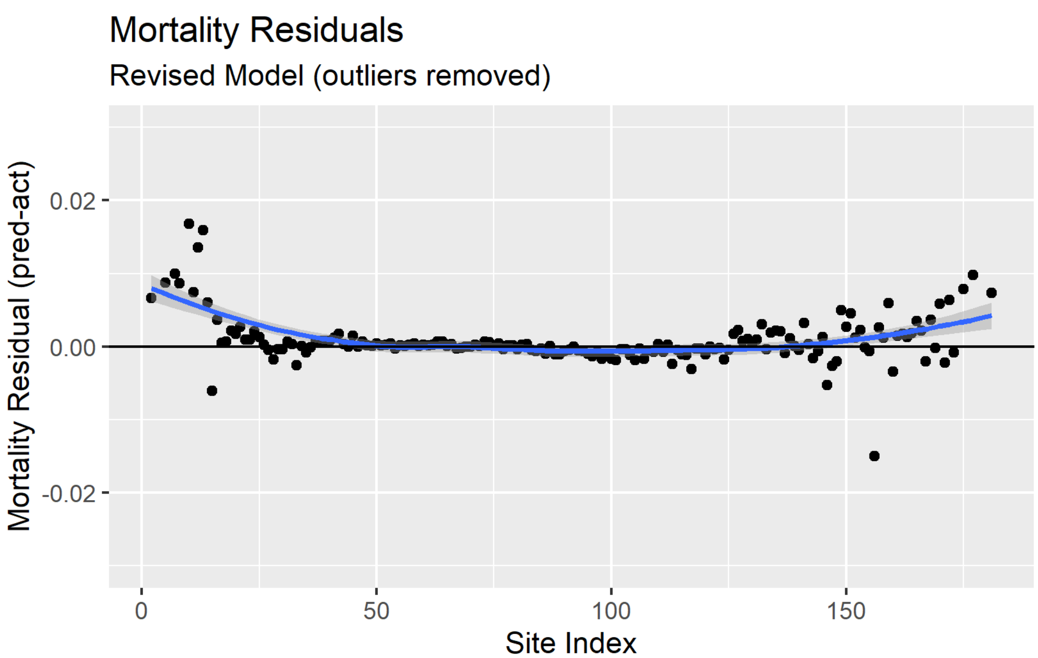
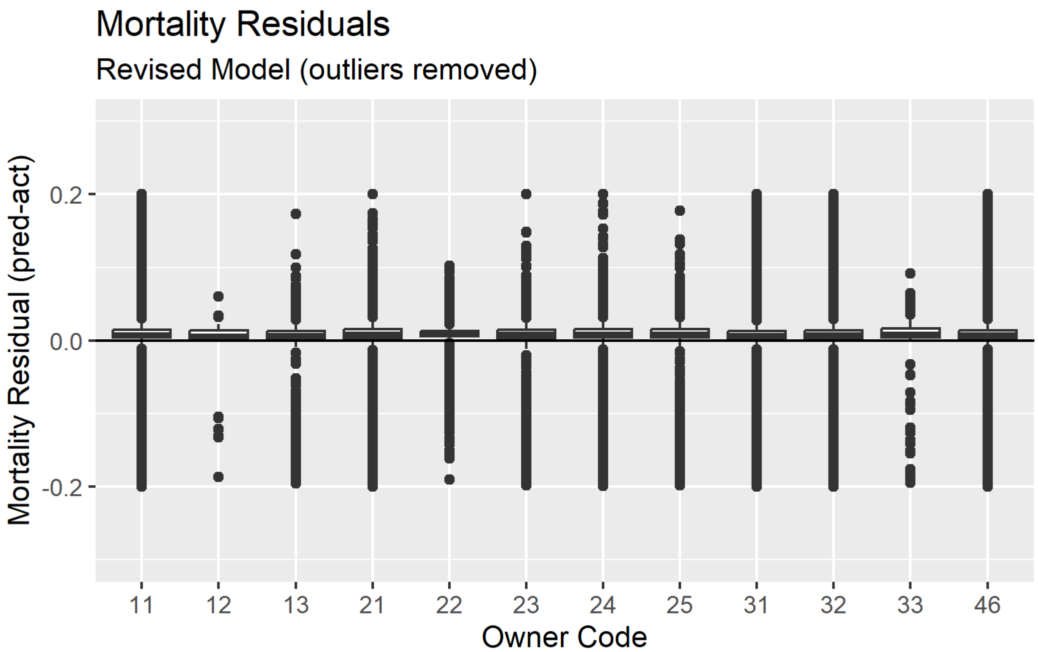
-0.02
-50
-40
-30
-20
-10
0
10
Extreme Minimum Temperature


Mortality Residuals
Revised Model (outliers removed)
Mortality Residual (pred-act)
0.02
0.00
-0.02
0
2500
5000
7500
10000
12500
Elevation (feet)


Mortality Residuals
Revised Model (outliers removed)
Mortality Residual (pred-act)
0.02
0.00
-0.02
0
200
400
600
Summer Precipitation


Mortality Residuals
Revised Model (outliers removed)
Mortality Residual (pred-act)
0.02
0.00
-0.02
0
50
100
150
Site Index


Mortality Residuals
Revised Model (outliers removed)
Mortality Residual (pred-act)
0.2
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-0.2
11
12
13
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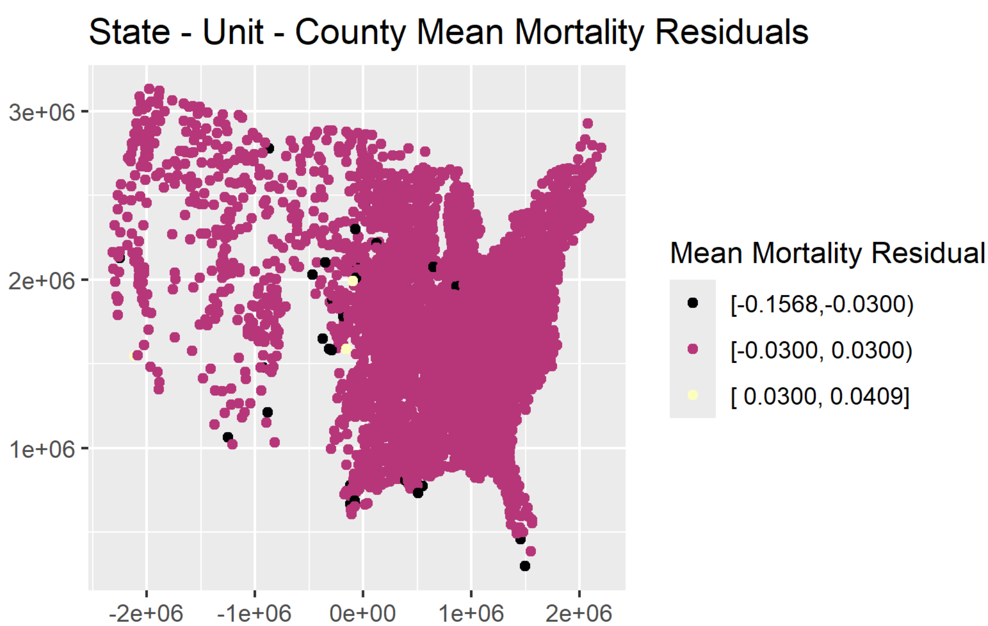
23
24
25
31
32
33
46
Owner Code


State - Unit - County Mean Mortality Residuals
3e+06
2e+06
1e+06
-2e+06
-1e+06
0e+00
1e+06
2e+06
Mean Mortality Residual
[-0.1568,-0.0300)
[-0.0300, 0.0300)
[0.0300, 0.0409]